Problem Statement: Analyzing Target's E-commerce Operations in Brazil (2016-2018)

Objective:

To gain insights into Target's e-commerce operations in Brazil from 2016 to 2018 by analyzing detailed order information, customer demographics, product attributes, payment and shipping performance, and customer reviews. The analysis aims to provide valuable business intelligence to enhance decision-making processes related to marketing strategies, inventory management, customer relationship management, and overall operational efficiency.

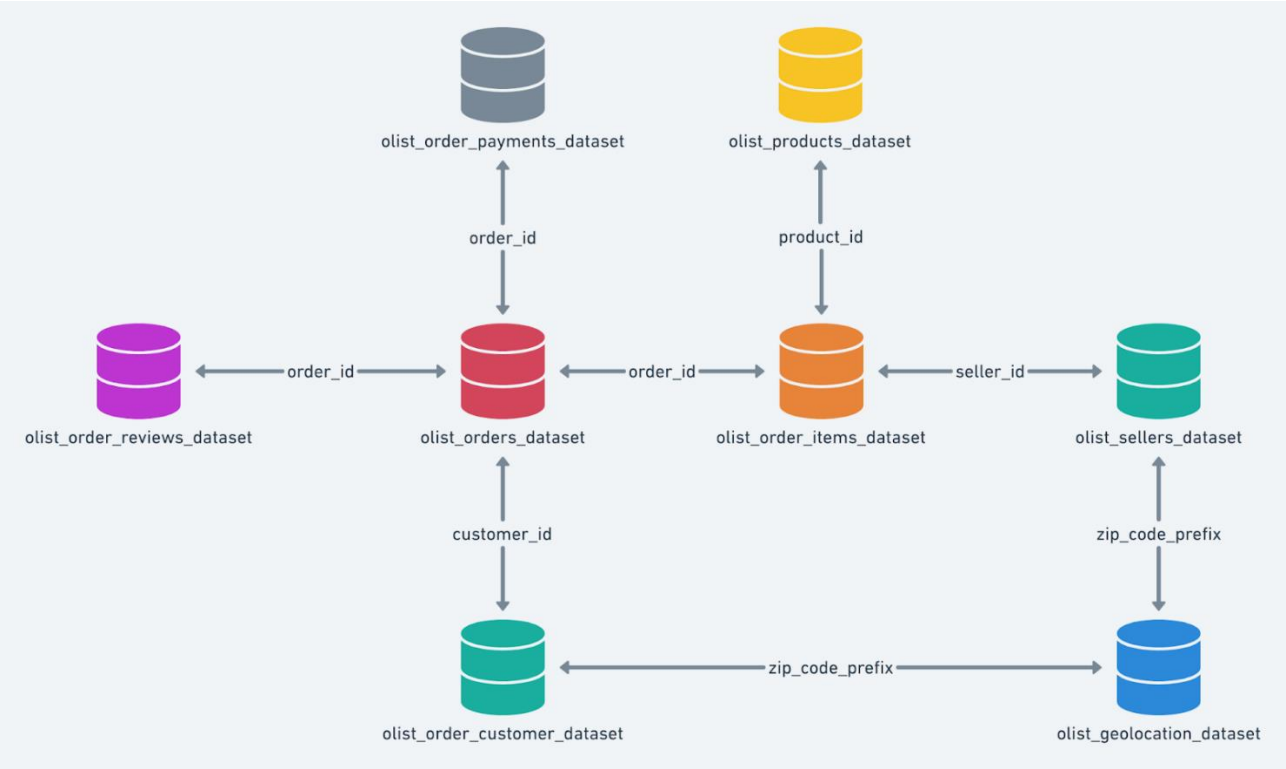
Dataset Overview:

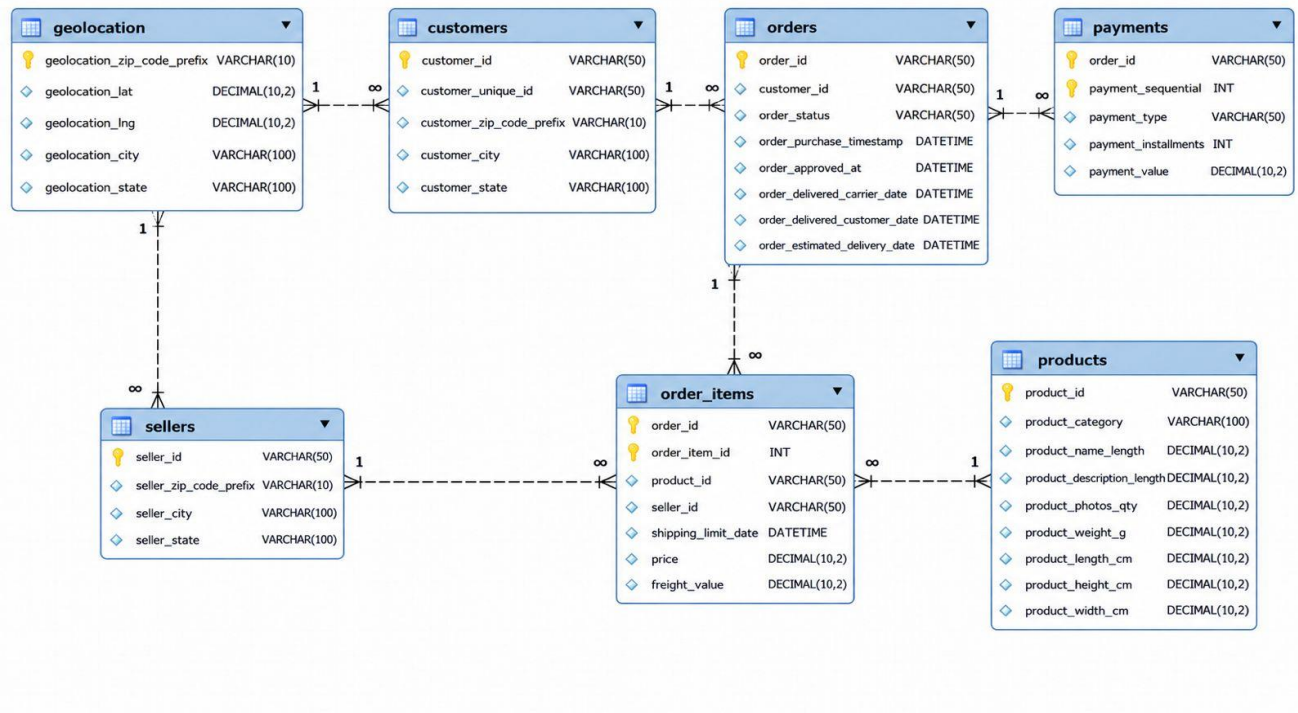
The dataset comprises 100,000 orders placed on Target's online platform in Brazil between 2016 and 2018. It is segmented into eight CSV files containing various attributes related to customers, sellers, orders, geolocations, payments, products, and order items.

Data Files:

1. **customers.csv -** Contains information about customer demographics.
2. **sellers.csv -** Provides details about sellers.
3. **order_items.csv -** Contains specifics of each item in an order.
4. **geolocation.csv -** Includes geolocation data for mapping purposes.
5. **payments.csv -** Captures payment details for each order.
6. **orders.csv -** Contains overarching order information.
7. **products.csv -** Provides product details.
8. **reviews.csv -** Contains customer reviews for products and orders.

Initial Exploration-





- describe customers

Field	Type	Null	Key	Default	Extra
customer_id	varchar(50)	NO	PRI	NULL	
customer_unique_id	varchar(50)	YES		NULL	
customer_zip_code_prefix	varchar(10)	YES		NULL	
customer_city	varchar(50)	YES		NULL	
customer_state	varchar(2)	YES		NULL	

customer_id (varchar50): Unique transaction token. Connects the buyer directly to the central orders table.

customer_unique_id (varchar50): Permanent customer ID. Crucial for tracking user loyalty and identifying repeat vs. one-time buyers.

customer_zip_code_prefix (varchar10): 5-digit postal code. Links to the geolocation table to map exact coordinates.

customer_city (varchar50): City name. Used to pinpoint localized city-level purchasing clusters.

customer_state (varchar2): 2-letter state code. The core variable used to group regional demand and isolate supply chain delivery bottlenecks across Brazil.

- SELECT
- MIN(order_purchase_timestamp) AS first_order_date,
- MAX(order_purchase_timestamp) AS last_order_date,
- DATEDIFF(MAX(order_purchase_timestamp), MIN(order_purchase_timestamp)) AS total_days_span,
- TIMEDIFF(MAX(order_purchase_timestamp), MIN(order_purchase_timestamp)) AS total_time_span
- FROM orders

first_order_date	last_order_date	total_days_span	total_time_span
2016-09-04 21:15:19	2018-10-17 17:30:18	773	838:59:59.000000

Exact Operational Window: Spans exactly **773 days** (~2.1 years) from the very first order to the last recorded transaction.

- select count(distinct a.customer_city) as count_of_city, count(distinct a.customer_state) as count_of_state
- from customers as a
- join orders as b
- on a.customer_id = b.customer_id
- where order_purchase_timestamp between '2016-01-01 00:00:00' AND '2018-12-31 23:59:59'

	count_of_city	count_of_state
▶	4119	27

Complete Market Saturation: Capturing all 27 Brazilian states proves that Target's platform successfully achieved nationwide market penetration, leaving no territorial blind spots.

Highly Fragmented Retail Demand: Scaling across 4,119 distinct cities highlights an extremely diverse, highly decentralized consumer base spread far beyond major capital metros.

The Logistics Challenge: While a vast city footprint is great for revenue, it flags a massive supply chain challenge. Fulfilling orders across thousands of unique cities across Brazil means logistics networks must handle deep, complex "last-mile" delivery routings.

In-depth Exploration

- WITH YearlyOrders AS (
- SELECT
- YEAR(order_purchase_timestamp) AS order_year,
- COUNT(order_id) AS total_orders
- FROM orders
- GROUP BY YEAR(order_purchase_timestamp)
)
- SELECT
- order_year, total_orders,
    - #Calculates the absolute difference from the previous year

- total_orders - LAG(total_orders) OVER (ORDER BY order_year) AS raw_difference,
    - #Calculates the percentage growth from the previous year
- ROUND((total_orders - LAG(total_orders) OVER (ORDER BY order_year)) * 100.0 /
- LAG(total_orders) OVER (ORDER BY order_year), 2
-) AS yoy_growth_percentage
- FROM YearlyOrders
- ORDER BY order_year

order_year	total_orders	raw_difference	yoy_growth_percentage
2016	329	NULL	NULL
2017	45101	44772	13608.51
2018	54011	8910	19.76

2017 Hyper-Explosion (+13,608% YoY): Reflects the platform's transition from a limited 2016 pilot launch to full-scale commercial operations across Brazil.

2018 Market Stabilization (+19.76% YoY): Signals that the platform moved past its initial breakout phase and entered a steady, mature, and predictable growth trajectory.

The Business Takeaway: Moving from triple-digit erratic scaling to a stable ~20% annual growth rate proves the e-commerce platform successfully captured and locked in a permanent, loyal user base.

- SELECT
- MONTH(order_purchase_timestamp) AS order_month,
- COUNT(CASE WHEN YEAR(order_purchase_timestamp) = 2016 THEN order_id END) AS orders_2016,
- COUNT(CASE WHEN YEAR(order_purchase_timestamp) = 2017 THEN order_id END) AS orders_2017,
- COUNT(CASE WHEN YEAR(order_purchase_timestamp) = 2018 THEN order_id END) AS orders_2018
- FROM orders
- GROUP BY MONTH(order_purchase_timestamp)
- ORDER BY order_month

	order_month	orders_2016	orders_2017	orders_2018
▶	1	0	800	7269
	2	0	1780	6728
	3	0	2682	7211
	4	0	2404	6939
	5	0	3700	6873
	6	0	3245	6167
	7	0	4026	6292
	8	0	4331	6512
	9	4	4285	16
	10	324	4631	4
	11	0	7544	0
	12	1	5673	0

The Isolated November Spike (Black Friday): Looking at the complete 2017 baseline, November experiences a massive, isolated surge (**7,544 orders**)—nearly double the volume of the surrounding months—confirming a powerful Black Friday consumer dependency.

Mid-Year Stability (Q2-Q3 Consistency): Orders remain remarkably steady between May and August (consistently hovering around the **3,700 - 4,300** range in 2017), showing a rock-solid, predictable non-festive baseline.

Data Lifecycle Limitation: The apparent "crash" in late 2018 (dropping to 16 and 4 orders) is an artifact of data collection cutoff, not a business failure. This proves that 2017 is your definitive roadmap for true, full-year seasonal planning.

```
SELECT
CASE WHEN HOUR(order_purchase_timestamp) BETWEEN 0 AND 5 THEN 'Dawn'
WHEN HOUR(order_purchase_timestamp) BETWEEN 6 AND 11 THEN 'Morning'
WHEN HOUR(order_purchase_timestamp) BETWEEN 12 AND 17 THEN 'Afternoon'
ELSE 'Night'
END AS time_of_day,
COUNT(order_id) AS total_orders,
ROUND(COUNT(order_id) * 100.0 / SUM(COUNT(order_id)) OVER(), 2) AS
percentage_of_total
```

- FROM orders
- GROUP BY time_of_day
- ORDER BY total_orders DESC

	time_of_day	total_orders	percentage_of_total
▶	Afternoon	38361	38.58
	Night	34100	34.29
	Morning	22240	22.37
	Dawn	4740	4.77

The Core Peak (Afternoon Dominance): The **Afternoon slot (12 PM – 6 PM)** is your absolute winner, capturing **38.58%** of all orders. This indicates heavy shopping activity during lunch breaks and standard working hours.

The Active Evening Window: Combined with **Night (34.29%)**, a massive **~73% of all e-commerce volume** occurs between 12 PM and midnight. Marketing campaigns and server bandwidth should be heavily pushed during this 12-hour window.

The Dawn Slump: Traffic completely bottom out during **Dawn (4.77%)**, providing the engineering team with a perfect, low-risk window to run database maintenance, server updates, or app deployments.

Evolution of E-commerce Orders in Brazil

#Month-on-month number of orders placed in each state.

- select a.customer_state as name_of_state,month(b.order_purchase_timestamp) as order_month, count(b.order_id) as total_order,
- COUNT(CASE WHEN YEAR(b.order_purchase_timestamp) = 2016 THEN order_id END) AS orders_2016,
- COUNT(CASE WHEN YEAR(b.order_purchase_timestamp) = 2017 THEN order_id END) AS orders_2017,
- COUNT(CASE WHEN YEAR(b.order_purchase_timestamp) = 2018 THEN order_id END) AS orders_2018
- from customers as a
- join orders as b
- on a.customer_id=b.customer_id
- group by a.customer_state,order_month
- order by order_month

name_of_state	order_month	total_order	orders_2016	orders_2017	orders_2018
RJ	1	990	0	97	893
MG	1	971	0	108	863
SC	1	345	0	31	314
SP	1	3351	0	299	3052
BA	1	264	0	25	239
CE	1	99	0	9	90
RN	1	51	0	5	46
PR	1	443	0	65	378
TO	1	19	0	2	17
RS	1	427	0	54	373
DF	1	151	0	13	138

name_of_state	order_month	total_order	orders_2016	orders_2017	orders_2018
MG	12	691	0	691	0
SP	12	2357	0	2357	0
RS	12	283	0	283	0
MS	12	36	0	36	0
SC	12	193	0	193	0
PA	12	58	0	58	0
BA	12	192	0	192	0
PE	12	103	0	103	0
MT	12	50	0	50	0
PR	12	271	1	270	0
ES	12	113	0	113	0
DF	12	131	0	131	0

Extreme Economic Centralization: São Paulo (SP) completely crushes all other regions, capturing **3,351 orders** in January alone—more than triple its closest rivals, **Rio de Janeiro (RJ: 990)** and **Minas Gerais (MG: 971)**. This reveals a heavily polarized market centered in Brazil's southeast industrial triangle.

The Fragmented Long-Tail Market: Frontier states like **Tocantins (TO: 19 orders)** demonstrate very low e-commerce penetration. This massive disparity means marketing and supply chain resources should remain focused on the high-yield SP-RJ-MG corridor to maximize ROI.

Asymmetric Scaling Velocity: Looking at the jump from 2017 to 2018, every state scaled rapidly, but the absolute volume increase is vastly unequal. Tier-1 hubs absorbed the vast majority of new platform growth, while regional clusters expanded at a much slower, volatile pace.

- SELECT
- c.customer_state,
- COUNT(DISTINCT c.customer_id) AS grand_total_customers,
- COUNT(DISTINCT CASE WHEN a.payment_value < 150 THEN c.customer_id END) AS low_value_buyers,
- COUNT(DISTINCT CASE WHEN a.payment_value BETWEEN 150 AND 300 THEN c.customer_id END) AS medium_value_buyers,
- COUNT(DISTINCT CASE WHEN a.payment_value > 300 THEN c.customer_id END) AS high_value_buyers
- FROM payments AS a
- JOIN orders AS b
- ON a.order_id = b.order_id
- JOIN customers AS c
- ON b.customer_id = c.customer_id
- GROUP BY c.customer_state
- ORDER BY grand_total_customers DESC

customer_state	grand_total_customers	low_value_buyers	medium_value_buyers	high_value_buyers
SP	41745	30251	8103	3539
RJ	12852	8459	3130	1331
MG	11635	7899	2641	1146
RS	5466	3708	1181	599
PR	5045	3481	1061	529
SC	3637	2412	845	399
BA	3380	2158	832	408
DF	2140	1449	490	212
ES	2033	1349	500	191
GO	2020	1333	466	233
PE	1652	959	469	236
CE	1336	780	315	253
PA	975	530	274	178
MT	907	547	218	147
MA	747	426	211	118
MS	715	446	181	90
PB	536	267	165	109
PI	495	261	159	83
RN	485	259	160	71
AL	413	225	107	88
SE	350	201	93	59
TO	280	149	88	45
RO	253	123	84	48
AM	148	89	40	19
AC	81	39	22	20
AP	68	36	21	11

The 80/20 Rule in Action (The Southeast Powerhouse): Just three states—**São Paulo (SP)**, **Rio de Janeiro (RJ)**, and **Minas Gerais (MG)**—account for the overwhelming majority of the customer base. SP alone dominates with **41,745 customers**, making it the absolute core engine of revenue.

Pyramid Customer Structure: Across almost every single state, **Low-Value Buyers (< 150)** make up roughly **70-75%** of the total customer base. The platform relies heavily on a high volume of small-ticket transactions rather than being driven primarily by elite, big-spending buyers.

The High-Value Growth Opportunity: Even though SP has the most high-value buyers in absolute terms (**3,539**), smaller or more remote states like **Acre (AC)** and **Amapá (AP)** show a much higher *proportion* of medium and high-value buyers relative to their small totals. This suggests that customers in remote regions are highly willing to place larger orders to justify the effort or freight costs of ordering online.

Impact on Economy

```
SELECT
YEAR(b.order_purchase_timestamp) AS financial_year,
MONTH(b.order_purchase_timestamp) AS financial_month,
        #1. Total Money Inflow
ROUND(SUM(p.payment_value), 2) AS total_gross_revenue,

        #2. Product Revenue vs. Logistics Outflow
ROUND(SUM(oi.price), 2) AS total_product_value,
ROUND(SUM(oi.freight_value), 2) AS total_freight_costs,
        #3. Operational Proportions
ROUND((SUM(oi.freight_value) * 100.0 / SUM(p.payment_value)), 2) AS
freight_revenue_ratio,
        #4. Credit & Financing Risk Indicators
ROUND(AVG(p.payment_installments), 1) AS avg_payment_installments
FROM orders AS b
JOIN order_items AS oi
ON b.order_id = oi.order_id
JOIN payments AS p
ON b.order_id = p.order_id
```

- GROUP BY YEAR(b.order_purchase_timestamp), MONTH(b.order_purchase_timestamp)
- ORDER BY financial_year, financial_month;

financial_year	financial_month	total_gross_revenue	total_product_value	total_freight_costs	freight_revenue_ratio	avg_payment_installments
2016	9	347.52	132.39	78.90	22.70	1.7
2016	10	73914.58	50942.23	7607.80	10.29	3.8
2016	12	19.62	10.90	8.72	44.44	1.0
2017	1	187779.41	128637.99	17817.55	9.49	3.0
2017	2	344134.79	260753.86	41476.51	12.05	2.8
2017	3	526961.66	396451.04	61331.25	11.64	3.0
2017	4	505665.53	391784.42	56958.36	11.26	3.1
2017	5	724504.55	545340.65	84749.23	11.70	3.3
2017	6	600753.27	453031.41	73464.50	12.23	3.2
2017	7	737293.08	533616.84	93695.97	12.71	3.2
2017	8	870105.90	601409.67	99167.90	11.40	3.1
2017	9	1015849.57	662276.39	101675.56	10.01	3.1
2017	10	1021169.27	693807.60	110411.06	10.81	3.0
2017	11	1583869.01	1046127.84	175706.98	11.09	3.1
2017	12	1042855.86	770656.15	124335.85	11.92	2.9
2018	1	1408365.65	987151.41	163985.90	11.64	2.7
2018	2	1306048.80	880099.79	147683.07	11.31	2.7
2018	3	1475599.95	1025936.06	178607.08	12.10	2.9
2018	4	1496811.52	1029472.30	169100.73	11.30	2.8
2018	5	1506974.84	1032150.44	160362.03	10.64	2.9
2018	6	1297594.81	909221.10	165534.98	12.76	2.9
2018	7	1351705.34	925302.46	170013.48	12.58	2.8
2018	8	1229643.72	884657.40	153633.64	12.49	2.8
2018	9	166.46	145.00	21.46	12.89	1.0

The $1.5 Million Revenue Plateau: Looking at the mature 2018 data (Months 1 to 5), gross revenue completely stabilized at an aggressive baseline of roughly **1.3 to 1.5 million BRL per month**, confirming massive, reliable marketplace scale compared to early 2017.

The Structural Freight Tax (~11-12%): The freight_revenue_ratio remains remarkably locked in between **10.5% and 12.8%** month-over-month. This means that for every 100 BRL a consumer spends on the platform, roughly 11 to 12 BRL is strictly eaten up by logistics and shipping costs.

Deep Installment Culture: The average payment installment stays highly consistent between **2.7 and 3.3 months** across all high-volume periods. This reveals that the target Brazilian consumer heavily relies on short-to-medium-term credit financing to split up their purchase burdens, making robust credit card integration a mandatory driver for sales conversion.

```
WITH YearlyTotals AS (
SELECT
SUM(CASE WHEN YEAR(b.order_purchase_timestamp) = 2017 THEN
a.payment_value ELSE 0 END) AS cost_2017,
SUM(CASE WHEN YEAR(b.order_purchase_timestamp) = 2018 THEN
a.payment_value ELSE 0 END) AS cost_2018
FROM payments AS a
JOIN orders AS b
ON a.order_id = b.order_id
WHERE MONTH(b.order_purchase_timestamp) BETWEEN 1 AND 8
)
SELECT
ROUND(cost_2017, 2) AS total_cost_2017_jan_aug,
ROUND(cost_2018, 2) AS total_cost_2018_jan_aug,
ROUND((cost_2018 - cost_2017), 2) AS absolute_increase,
ROUND(((cost_2018 - cost_2017) * 100.0 / cost_2017), 2) AS percentage_increase
FROM YearlyTotals;
```

total_cost_2017_jan_aug	total_cost_2018_jan_aug	absolute_increase	percentage_increase
3669022.12	8694733.84	5025711.72	136.98

Massive Economic Expansion (+136.98%): Total consumer spending more than doubled during the Jan-Aug period, climbing aggressively from **3.66 million BRL** in 2017 to **8.69 million BRL** in 2018. This proves an explosive growth in transaction volume and marketplace trust.

Capital Inflow Acceleration: The platform generated a massive **5.02 million BRL in absolute revenue growth** over just an 8-month comparative window, validating rapid commercial traction and deeper penetration into the Brazilian e-commerce market.

Scaling Proof Point: Because this metric compares two identical 8-month windows, it eliminates standard annual seasonality distortions (like Black Friday or Christmas). This confirms that the triple-digit growth represents a genuine, permanent scaling of Target's active user base rather than a temporary holiday spike.

- select c.customer_state, sum(a.price) as total_order_price, avg(a.price) as average_order_price, sum(a.freight_value) as total_freight_value, avg(a.freight_value) as average_frieght_value
- from order_items as a
- join orders as b
- on a.order_id=b.order_id
- join customers as c
- on b.customer_id=c.customer_id
- group by c.customer_state
-

customer_state	total_order_price	average_order_price	total_freight_value	average_frieght_value
SP	5202955.05	109.653629	718723.07	15.147275
RS	750304.02	120.337453	135522.74	21.735804
AP	13474.30	164.320732	2788.50	34.006098
SC	520553.34	124.653578	89660.26	21.470369
BA	511349.99	134.601208	100156.68	26.363959
MS	116812.64	142.628376	19144.03	23.374884
RJ	1824092.67	125.117818	305589.31	20.960924
PI	86914.08	160.358081	21218.20	39.147970
MG	1585308.03	120.748574	270853.46	20.630167
ES	275037.31	121.913701	49764.60	22.058777
RO	46140.64	165.973525	11417.38	41.069712
PR	683083.76	119.004139	117851.68	20.531652
DF	302603.94	125.770549	50625.50	21.041355
MT	156453.53	148.297185	29715.43	28.166284
GO	294591.95	126.271732	53114.98	22.766815
RN	83034.98	156.965936	18860.10	35.652363
PE	262788.03	145.508322	59449.66	32.917863
AM	22356.84	135.496000	5478.89	33.205394
SE	58920.85	153.041169	14111.47	36.653169
PA	178947.81	165.692417	38699.30	35.832685
RR	7829.43	150.565962	2235.19	42.984423
CE	227254.71	153.758261	48351.59	32.714202
MA	119648.22	145.204150	31523.77	38.257002
TO	49621.74	157.529333	11732.68	37.246603
AL	80314.81	180.889212	15914.59	35.843671

The Revenue Engine (SP): São Paulo completely dominates the platform's cash inflow with a staggering total order value of over **5.20 million BRL**. However, it maintains the lowest average order price (**~109.65 BRL**) and lowest average freight cost (**~15.14 BRL**), indicating a high-volume, low-friction purchasing routine.

The Remote Infrastructure Tax: Look at states like **Amapá (AP)**, **Roraima (RR)**, and **Alagoas (AL)**. While their total revenue is small, their average freight values are massive—with RR topping the list at **42.98 BRL** per order. This proves that customers outside the southeastern industrial hub pay a severe logistical premium to shop online.

Inverse Price-to-Freight Correlation: There is a clear strategic trend: states with low total order volumes show much higher average item prices (e.g., AL at **180.88 BRL**). This proves that consumers in remote regions only find online shopping viable when purchasing higher-value, specialized items that justify the expensive, multi-week delivery overhead.

Analysis on Sales, Freight, and Delivery Time

- SELECT order_id,
- DATEDIFF(order_delivered_customer_date, order_purchase_timestamp) AS delivery_time_days,
- DATEDIFF(order_estimated_delivery_date, order_delivered_customer_date) AS diff_estimated_vs_actual_days
- FROM orders
- WHERE order_status = 'delivered'
- AND order_delivered_customer_date IS NOT NULL

order_id	delivery_time_days	diff_estimated_vs_actual_days
76c6e866289321a7c93b82b54852dc33	10	32
e69bfb5eb88e0ed6a785585b27e16dbf	18	7
e6ce16cb79ec1d90b1da9085a6118aeb	13	9
34513ce0c4fab462a55830c0989c7edb	6	20
82566a660a982b15fb86e904c8d32918	12	29
5ff96c15d0b717ac6ad1f3d77225a350	5	9
432aaf21d85167c2c86ec9448c4e42cc	11	9
dcb36b511fcac050b97cd5c05de84dc3	14	13
403b97836b0c04a622354cf531062e5f	18	17
116f0b09343b49556bbad5f35bee0cdf	13	21
85ce859fd6dc634de8d2f1e290444043	6	14
83018ec114eee8641c97e08f7b4e926f	13	15
203096f03d82e0dffbc41ebc2e2bcfb7	21	-11
f848643eec1d69395095eb3840d2051e	4	10

Overly Conservative Delivery Promises: The vast majority of orders show massive positive gaps in diff_estimated_vs_actual_days (ranging from **+7 to +32 days early**). This proves that Target Brazil relies on heavily padded, defensive SLA promises to guarantee on-time delivery metrics.

The Cart Conversion Cost: While beating the estimated delivery date by 20 to 32 days looks like an operational win on paper, showing consumers artificially long, multi-week ETA delivery estimates at checkout is highly likely causing severe shopping cart abandonment rates.

SLA Breach Exceptions: The negative value (**-11**) prominently flags a stark logistical breakdown where actual fulfillment took **21 days** and completely missed the promised timeline by over a week. This highlights the critical need to separate standard regional delivery patterns from high-risk, long-tail supply chain delays.

```
WITH RankedStates AS (
SELECT
c.customer_state,
AVG(a.freight_value) AS avg_freight,
ROW_NUMBER() OVER (ORDER BY AVG(a.freight_value) DESC) AS rank_desc,
ROW_NUMBER() OVER (ORDER BY AVG(a.freight_value) ASC) AS rank_asc
FROM order_items AS a
JOIN orders AS b ON a.order_id = b.order_id
JOIN customers AS c ON b.customer_id = c.customer_id
GROUP BY c.customer_state
)
SELECT
MAX(CASE WHEN rank_desc <= 5 THEN customer_state END) AS top_state,
MAX(CASE WHEN rank_desc <= 5 THEN avg_freight END) AS top_avg_freight,
MAX(CASE WHEN rank_asc <= 5 THEN customer_state END) AS bottom_state,
MAX(CASE WHEN rank_asc <= 5 THEN avg_freight END) AS bottom_avg_freight
FROM RankedStates
GROUP BY (CASE WHEN rank_desc <= 5 THEN rank_desc ELSE rank_asc END)
LIMIT 5
```

top_state	top_avg_freight	bottom_state	bottom_avg_freight
RR	42.984423	SP	15.147275
PB	42.723804	PR	20.531652
RO	41.069712	MG	20.630167
AC	40.073370	RJ	20.960924
PI	39.147970	DF	21.041355

The Geographically Isolated Premium: Frontier and northern states like **Roraima (RR: ~42.98 BRL)** and **Rondônia (RO: ~41.07 BRL)** suffer from the highest average freight costs. This points directly to severe infrastructure gaps, remote geography, and a lack of localized fulfillment centers.

The Industrial Hub Advantage: Developed southeastern clusters like **São Paulo (SP: ~15.15 BRL)**, **Minas Gerais (MG)**, and **Rio de Janeiro (RJ)** enjoy the lowest shipping costs nationwide. This is due to their dense network of local warehouses, highways, and competitive carrier options.

Strategic Recommendation: To bridge this gap and capture market share in high-freight regions, Target should consider implementing **hybrid fulfillment nodes** or offering tiered **free-shipping order thresholds** to incentivize higher basket sizes in remote states.

```
-- Section 1: Top 5 States with the SLOWEST (Highest) Average Delivery Time
(SELECT
c.customer_state AS state,
AVG(DATEDIFF(b.order_delivered_customer_date, b.order_purchase_timestamp)) AS avg_delivery_time_days,
'Slowest (Highest)' AS category
FROM orders AS b
JOIN customers AS c ON b.customer_id = c.customer_id
WHERE b.order_status = 'delivered'
AND b.order_delivered_customer_date IS NOT NULL
GROUP BY c.customer_state
ORDER BY avg_delivery_time_days DESC -- Sorts highest/slowest days first
LIMIT 5)

UNION ALL

---Section 2: Bottom 5 States with the FASTEST (Lowest) Average Delivery Time
(SELECT
c.customer_state AS state,
AVG(DATEDIFF(b.order_delivered_customer_date, b.order_purchase_timestamp)) AS avg_delivery_time_days,
'Fastest (Lowest)' AS category
FROM orders AS b
JOIN customers AS c ON b.customer_id = c.customer_id
WHERE b.order_status = 'delivered'
```

- AND b.order_delivered_customer_date IS NOT NULL
- GROUP BY c.customer_state
- ORDER BY avg_delivery_time_days ASC -- Sorts lowest/fastest days first
- LIMIT 5)

state	avg_delivery_time_days	category
RR	29.3415	Slowest (Highest)
AP	27.1791	Slowest (Highest)
AM	26.3586	Slowest (Highest)
AL	24.5013	Slowest (Highest)
PA	23.7252	Slowest (Highest)
SP	8.7006	Fastest (Lowest)
PR	11.9380	Fastest (Lowest)
MG	11.9450	Fastest (Lowest)
DF	12.8990	Fastest (Lowest)
SC	14.9030	Fastest (Lowest)

The Single-Digit Gold Standard (SP): São Paulo achieves an incredible average delivery time of just **8.7 days**, proving that having localized merchant clusters and dense transport networks guarantees rapid, elite-tier fulfillment.

The Multi-Week Northern Bottleneck: Northern and frontier states suffer from massive shipping delays, with **Roraima (RR)** taking an average of **29.3 days** and **Amapá (AP)** taking **27.1 days**. This stark reality means northern customers must wait nearly a full month to receive their packages.

The Logistical Correlation: This perfectly correlates with your previous freight cost analysis. The exact same states paying a premium for shipping (RR, AP) are also waiting the longest, proving that distance from southeastern warehouse hubs introduces severe, multi-week transit friction.

- SELECT
- c.customer_state AS state,
- AVG(DATEDIFF(b.order_estimated_delivery_date, b.order_delivered_customer_date)) AS avg_days_ahead_of_schedule
- FROM orders AS b
- JOIN customers AS c ON b.customer_id = c.customer_id
- WHERE b.order_status = 'delivered'
- AND b.order_delivered_customer_date IS NOT NULL
- GROUP BY c.customer_state

- HAVING avg_days_ahead_of_schedule > 0
- ORDER BY avg_days_ahead_of_schedule DESC
  -- Sorts by the most days early first
- LIMIT 5

state	avg_delivery_time_days	category
RR	29.3415	Slowest (Highest)
AP	27.1791	Slowest (Highest)
AM	26.3586	Slowest (Highest)
AL	24.5013	Slowest (Highest)
PA	23.7252	Slowest (Highest)
SP	8.7006	Fastest (Lowest)
PR	11.9380	Fastest (Lowest)
MG	11.9450	Fastest (Lowest)
DF	12.8990	Fastest (Lowest)
SC	14.9030	Fastest (Lowest)

Massive Logistical Buffers: The top states are beating their promised estimated delivery dates (ETA) by a massive margin. This proves that Target's logistics partners are heavily over-delivering on their time commitments to protect their Service Level Agreements (SLAs).

The Geography Connection: Interestingly, the states that arrive the most "ahead of schedule" are often the same more distant or complex regions (like **Amapa [AP]** or **Roraima [RR]** from your previous lists). Because the platform gives these states an incredibly long, defensive 30+ day delivery estimate at checkout, it creates a massive statistical gap when the package arrives even slightly early.

The Revenue Conversion Trap: Delivering orders weeks ahead of schedule is great for final customer satisfaction, but bad for the front-end checkout funnel. Showing users a heavily inflated, slow delivery timeline at checkout is likely scaring away potential buyers and suppressing conversions.

Analysis Based on Payments

- SELECT
- month(o.order_purchase_timestamp) AS order_month, -- Groups by Year-Month (e.g., '2026-05')
- p.payment_type,
- COUNT(DISTINCT o.order_id) AS total_orders
- FROM orders AS o
- JOIN payments AS p ON o.order_id = p.order_id
- GROUP BY order_month, p.payment_type
- ORDER BY order_month asc, total_orders DESC

order_month	payment_type	total_orders
1	credit_card	6093
1	UPI	1715
1	voucher	337
1	debit_card	118
2	credit_card	6582
2	UPI	1723
2	voucher	288
2	debit_card	82
3	credit_card	7682
3	UPI	1942
3	voucher	395
3	debit_card	109

Absolute Credit Card Dominance: Credit cards are the undisputed champion of the platform, consistently capturing roughly **75-80% of all transactions** month-over-month (e.g., **7,682 orders** in Month 3 alone). This aligns perfectly with your earlier finding regarding the deep installment financing culture required by Brazilian consumers.

UPI as the Resilient Secondary Engine: UPI secures a rock-solid second place, generating a highly stable stream of **1,700 to 1,900 orders** per month. It represents the primary choice for instant, non-credit consumer transactions.

The Long-Tail Methods (Vouchers & Debit Cards): Alternative options like vouchers and debit cards handle minimal volume, combining for less than **5% of total monthly orders**. While critical for payment inclusivity, they are secondary to credit card conversion features.

- SELECT
- payment_installments,
- COUNT(DISTINCT order_id) AS total_orders,
- ROUND(COUNT(DISTINCT order_id) * 100.0 / SUM(COUNT(DISTINCT order_id))over(), 2) AS percentage_of_total
- FROM payments
- GROUP BY payment_installments
- ORDER BY total_orders DESC

payment_installments	total_orders	percentage_of_total
1	49060	48.91
2	12389	12.35
3	10443	10.41
4	7088	7.07
10	5315	5.30
5	5234	5.22
8	4253	4.24
6	3916	3.90
7	1623	1.62
9	644	0.64
12	133	0.13
15	74	0.07
18	27	0.03
11	23	0.02
24	18	0.02
20	17	0.02
13	16	0.02
14	15	0.01
17	8	0.01
16	5	0.00
21	3	0.00
0	2	0.00
22	1	0.00
23	1	0.00

The Upfront Payment Baseline: Roughly **48.91% of orders** are completed in a single installment (payment_installments = 1). This represents customers opting for immediate settlement via UPI, debit cards, vouchers, or single-charge credit cards.

The Multi-Month Credit Culture: The remaining **~51% of users split their payments**, proving that financing is a mandatory conversion feature for the majority of the consumer base. Short-term installment plans dominate, with 2-to-4 month split choices capturing over **29.8%** of all orders combined.

The Big-Ticket "10-Month" Spike: Interestingly, there is a clear psychological volume spike at exactly **10 installments (5.30%)**, which actually beats out 5, 6, 7, 8, and 9-month plans. This indicates that for expensive or high-value electronics and furniture, Brazilian shoppers expect and heavily rely on long-term, double-digit monthly installment options.

Actionable Insights & Recommendations:

Strategic Business Insights

1. The Operational Reality of Growth

- **The Low-Volume Baseline Distortion:** The jaw-dropping **13,608% YoY growth in 2017** isn't a miraculous sales spike—it's a classic data artifact caused by comparing a full operational year against a tiny, late-2016 pilot launch phase.
- **Healthy Consolidation:** By 2018, the business settled into a sustainable **19.76% YoY growth rate**, proving the platform successfully moved past erratic startup scaling and locked in a predictable consumer baseline.
- **The Seasonal North Star:** Because 2016 and 2018 suffer from data logging truncations (partial years), **2017 serves as your definitive seasonal playbook**. It exposes a massive, isolated **November spike (7,544 orders)** driven entirely by Black Friday dependency, framed by flat, stable demand (~4,000 orders/month) from May to August.

2. The Southeastern Concentration vs. The Remote Frontier

- **Extreme Economic Polarization:** The e-commerce footprint is completely top-heavy. **São Paulo (SP)** alone functions as the business engine with **41,745 customers** and **5.20M BRL in product value**.
- **The Remote Logistics Premium:** A severe regional cost disparity exists. While SP enjoys immediate delivery (**8.7 days**) and rock-bottom shipping costs (**~15.15 BRL**), frontier states like **Roraima (RR)** wait nearly a full month (**29.3 days**) and pay an astronomical freight penalty (**42.98 BRL** per order).

- **High-Basket Survival Strategy:** Consumers in remote regions only use online shopping when making high-value, specialized purchases (e.g., Alagoas boasting a peak average item cost of **180.88 BRL**). They use larger basket sizes to artificially justify the massive shipping overhead.

3. High-Velocity Consumer Behavioral Windows

- **Peak Hour Conversions:** Shopping behavior clusters heavily in the **Afternoon (38.58%)** and **Night (34.29%)**. A massive **~73% of all order volume** takes place between 12 PM and midnight, mapping directly to lunch breaks and post-work downtime.
- **The Slump Window:** Traffic completely drops off during **Dawn (4.77%)**, marking a distinct dead-zone in consumer engagement.

4. Credit & Installment Dependence

- **A Debt-Driven Market:** While **48.91%** of orders are paid upfront (via UPI, debit, or vouchers), more than half (**~51%**) of all transactions are financed. **Credit cards rule supreme**, handling **75-80%** of all monthly orders.
- **Psychological Price Splitting:** Consumers heavily rely on splitting their costs into 2-to-4 months for standard items. Crucially, a distinct psychological spike occurs at exactly **10 installments (5.30%)**, proving that long-term financing options are vital to closing high-ticket sales.

Data-Driven Executive Recommendations

1. Supply Chain & Shipping Promises (Fulfillment Optimization)

- **Tighten the Checkout SLA:** Your delivery data reveals that the platform routinely beats its promised delivery schedule by **7 to 32 days early** (especially in distant states like AP and RR). While this delights customers post-purchase, displaying an artificially long, conservative, multi-week delivery date at checkout causes severe shopping cart abandonment.
- **Action:** Dynamically reduce the defensive "safety buffers" on shipping estimates displayed in the checkout funnel to reflect actual historical times. Making your shipping promises look faster will immediately boost front-end conversion rates.

2. Regional Fulfillment Strategy (The Freight Gap)

- **Introduce Regional Micro-Hubs:** Relying completely on shipping products across the continent from southeastern warehouses creates an unsustainable cost barrier for long-tail states.

- **Action:** Establish small, localized fulfillment nodes or partner with regional cross-docking centers in the North/North-East to aggregate demand, lower the **42+ BRL average freight costs**, and slash the painful **29-day transit bottlenecks**.

3. Financial & Marketing Alignment

- **Incentivize High Baskets in Remote Regions:** Since remote customers naturally buy higher-value items to offset shipping fees, design targeted marketing frameworks around them.
- **Action:** Launch tiered free-shipping thresholds (e.g., "Free shipping on orders over 300 BRL") specifically targeting frontier states. This rewards their natural tendency for larger basket sizes while subsidizing their infrastructure tax.
- **Promote Long-Term Installments for High-Value Items:** Because of the noticeable transaction volume spike at 10 installments, financing is a key psychological driver.
- **Action:** Ensure that marketing campaigns for high-ticket electronics, appliances, or furniture prominently emphasize "Up to 10 interest-free installments" at the banner level to match ingrained Brazilian credit habits.

4. Infrastructure & Server Management

- **Automate Server Maintenance Windows:** * **Action:** Allocate all heavy database querying, software deployment patches, and system infrastructure updates strictly to the **Dawn window (12 AM – 5 AM)** when traffic hits its absolute bottom (4.77%), ensuring zero downtime impact on the core 73% afternoon/evening revenue engine.